

Mohammed Siddiq

AI ML Engineer • MLOps & NLP • Enterprise Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with a B.Tech in AI & Machine Learning and AWS Cloud Practitioner certification, driving enterprise system reliability at Tech Mahindra for a US-based client. Skilled across the full ML lifecycle - NLP, transformer fine-tuning, RAG systems, and MLOps - backed by practical AWS knowledge in compute, storage, security, and well-architected cloud design. Independently reduced LMS engagement drops by 30% through content personalization and grew client acquisition 40% via a solo-built full-stack product. Seeking impactful roles at the intersection of AI research and production systems.

EXPERIENCE

Technical Operations Engineer | Tech Mahindra

Aug 2025 – Present • Hyderabad, India

US-Based Enterprise Client Engagement

- Analyzed structured operational data using **SQL** to identify failure patterns, communication faults, and system issues, supporting data-driven troubleshooting and root-cause identification.
- Investigated production incidents and customer-reported issues, identifying defects and improvement opportunities while coordinating with cross-functional teams for resolution.
- Managed operational cases across a large-scale **2,000+ site enterprise environment**, prioritizing issues based on urgency and business impact.
- Maintained accurate technical documentation, incident records, configuration changes, and resolution details to ensure traceability and consistent operational support.

Machine Learning Intern | Ignitus

Feb 2024 – May 2024 • Remote

- Developed Supervised Machine Learning models and NLP solutions, performing data preprocessing, feature engineering, model development, testing, and deployment.
- Built reusable software components and collaborated through Git-based development workflows, including feature branches, pull requests, and peer reviews.
- Worked within an Agile team to deliver production-ready features while continuously improving code quality and application performance.

Web Developer (Freelance) | Excel Placement Services

Dec 2024 – Jan 2025 • Hyderabad, India

- Designed, developed, and deployed a full-stack responsive website, leading the complete project lifecycle from requirements gathering and UI development to production deployment, resulting in a 40% increase in client acquisition inquiries.
- Managed end-to-end website migration by migrating the codebase, configuring GoDaddy hosting, DNS records, domain setup, backend components, and production deployment, ensuring a seamless transition with minimal downtime.
- Optimized application performance by implementing cloud deployment and hosting best practices, reducing response latency from 1–1.5 seconds to under 500 ms while improving application reliability and user experience.
- Collaborated with stakeholders to translate business requirements into functional specifications, delivering iterative enhancements through continuous feedback and providing post-deployment support to ensure long-term application stability.

PROJECTS

Natural Language Question Answering System

- Built a Flask-based REST API enabling users to query unstructured PDF documents via transformer-powered NLP. Handled full ownership from model selection and API design to backend integration and deployment documentation.

Tech Stack: *Python • Flask • BERT • Hugging Face Transformers • HTML • CSS • JavaScript • REST API*

NASA Astronomy Picture of the Day (APOD)

- Developed a cloud-enabled web application that integrates with NASA's Astronomy Picture of the Day (APOD) API, allowing users to view daily images, search by date, and explore random astronomy content through a responsive interface.
- Implemented a serverless API proxy using AWS Lambda, Amazon API Gateway, CloudFront, and IAM to gain hands-on experience with cloud-native application architecture, API integration, and secure request routing while deploying the frontend on GitHub Pages.

Tech Stack: *HTML5, CSS3, JavaScript, REST API, AWS Lambda, Amazon API Gateway, Amazon CloudFront, AWS IAM, GitHub Pages*

RAG-Powered Multi-Document QA System

- Engineered an end-to-end Retrieval-Augmented Generation pipeline with multi-format document ingestion (PDF, DOCX, TXT), semantic vector storage via ChromaDB, and advanced retrieval strategies (MMR + MultiQueryRetriever), delivering cited, cross-document synthesised answers at scale using LLaMA 3.3 70B via Groq API with full source attribution and strategy-switching CLI.

Tech Stack: *Python • LangChain • ChromaDB • HuggingFace Sentence Transformers • Groq API • LLaMA 3.3 70B*

TECHNICAL SKILLS

Languages: Python, SQL

ML / AI: NLP, Deep Learning, Generative AI, LLMs, RAG Pipelines, Vector Embeddings, Prompt Engineering, Transfer Learning

Frameworks & Tools: Computer Vision, PyTorch, Keras, Hugging Face, LangChain, Pandas, NumPy, Flask

MLOps & Cloud: MLflow, DVC, Docker, GitHub Actions, AWS (EC2, S3, Lambda)

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

2020 – 2024

Siddhartha Institute of Technology and Science, JNTUH

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Database Management Systems, Computer Networks

CERTIFICATIONS

- **AWS** Cloud Practitioner Essentials – AWS Training & Certification (Jul 2026)
- Machine Learning Training – Internshala
- Deep Learning with TensorFlow – IBM
- Machine Learning with Python – freeCodeCamp
- Responsive Web Design – freeCodeCamp